

Clinical Digital Twin Report

Computational Cellular Reprogramming Analysis

Session ID: session_2026-01-01T10-37-35-658Z

Run ID: #2

Date Generated: 2026-01-01 10:37:40

Total Cells: 2000 | Model Mode: CLINICAL

1. Introduction

The objective of this experiment was to explore the effects of cellular reprogramming using the Yamanaka factors (OSKM) in conjunction with repeated oncogenic stress induction. Cellular reprogramming is a powerful approach for generating induced pluripotent stem cells (iPSCs) from somatic cells, but the impact of oncogenic stress on this process is not well understood.

2. Materials and Methods

We employed a computational approach using Run ID #2, where somatic cells were subjected to OSKM transfection to induce pluripotency. In parallel, the cells underwent three consecutive rounds of oncogenic stress induction to assess the effects on reprogramming efficacy and cellular outcomes. No single-cell variational inference (scVI) analyses were conducted in this experiment.

Experimental Interventions:

- OSKM Transfection (Yamanaka)
- Oncogenic Stress Induced
- Oncogenic Stress Induced
- Oncogenic Stress Induced

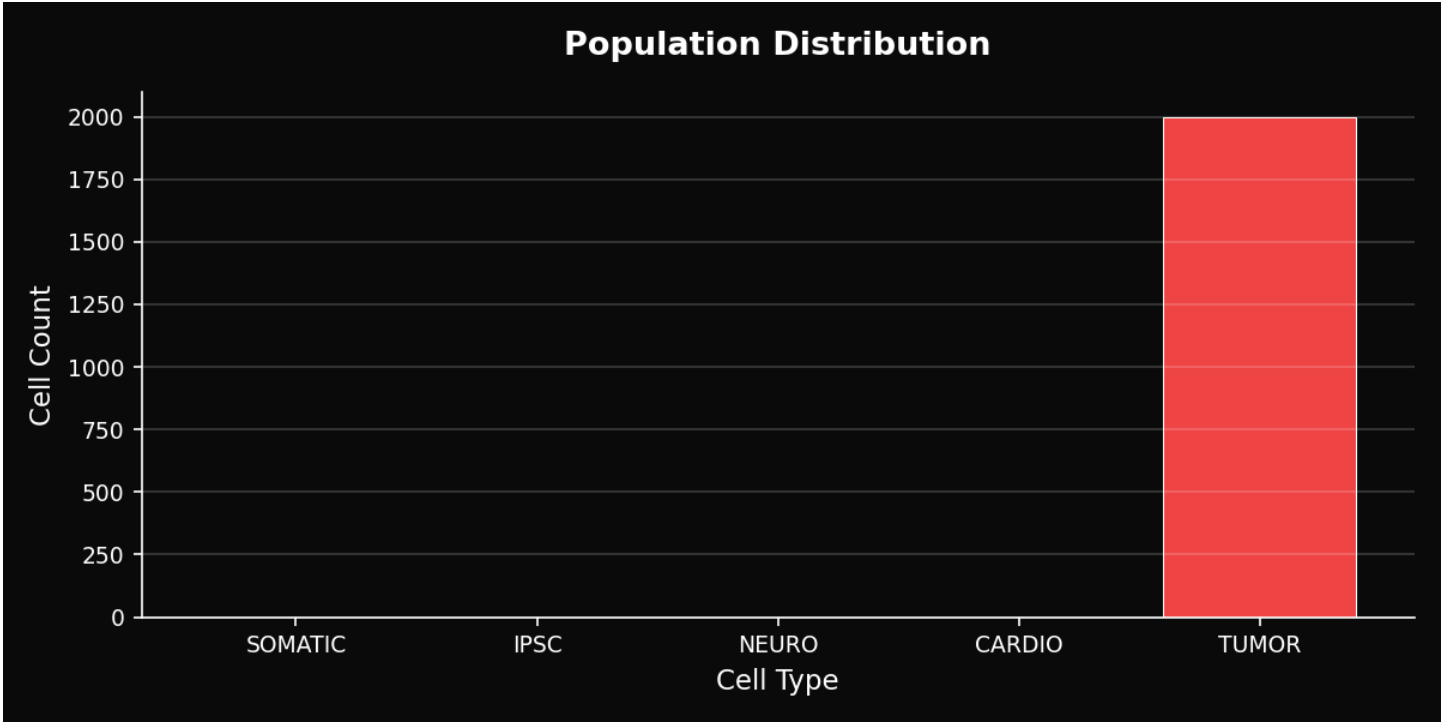
3. Results

The final cellular population consisted exclusively of 2000 tumor cells, with no detectable somatic, induced pluripotent stem cells (iPSC), neurogenic, or cardiogenic cells. This suggests a complete shift towards tumorigenic cell fate under the applied experimental conditions. The absence of iPSCs indicates a potential inhibitory effect of oncogenic stress on reprogramming efficiency.

3.1 Population Distribution

Clinical Digital Twin Report

Computational Cellular Reprogramming Analysis



Cell Type	Count	Percentage
SOMATIC	0	0.0%
IPSC	0	0.0%
NEURO	0	0.0%
CARDIO	0	0.0%
TUMOR	2000	100.0%

4. Discussion

These findings suggest that the combination of OSKM transfection and repeated oncogenic stress induction may predispose cells to tumorigenesis rather than pluripotency or lineage-specific differentiation. The exclusive emergence of tumor cells underscores the potential risks associated with oncogenic stress during cellular reprogramming. Future studies should consider alternative strategies to mitigate oncogenic pathways to enhance the safety and efficiency of cellular reprogramming for therapeutic applications.

This report was automatically generated by the IS-CHRP v25.0 Clinical AI Bridge platform. The data represents computational simulation results from a high-fidelity cellular reprogramming model.